

SPRING 2022

1. Discuss Nash equilibrium in respect of "Prisoners Dilemma".

In the Prisoner's Dilemma, two individuals (prisoners) are faced with the decision to cooperate or defect. If both cooperate, they both receive a moderate sentence. If one defects while the other cooperates, the defector receives a minimal sentence while the cooperator receives a maximal one. If both defect, they both receive a moderately severe sentence.

-Diagram

A payoff matrix can be used to illustrate the possible outcomes for each player based on their choices.

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		Player 2	
		Cooperate	Defect
Player 1	Cooperate	(-2, -2)	(-10, 0)
	Defect	(0, -10)	(-5, -5)

...

- Example: Suppose two criminals are arrested, and the police separate them, offering each a deal to testify against the other. Regardless of what the other chooses, each prisoner has a dominant strategy to defect, as defecting leads to a lower sentence regardless of the other's choice. This results in both prisoners defecting, even though cooperating would yield a better outcome for both.

2. **Global Risk Aversion and Measure of Relative Risk Aversion**:

Explanation: Global risk aversion refers to the overall sentiment or attitude towards risk across all financial markets worldwide. It reflects the general willingness of investors to accept risk in their investment decisions.

- Diagram: This concept is typically not represented with a diagram but rather observed through trends in financial markets and investment behavior on a global scale.

- Example: During times of economic uncertainty or geopolitical instability, global risk aversion tends to increase, leading investors to seek safer investments such as government bonds or gold, while avoiding riskier assets like stocks or commodities.

3. **Attributes of Public Goods:**

- Explanation: Public goods have two main attributes: non-excludability and non-rivalrous consumption. Non-excludability means that individuals cannot be excluded from using the good once it is provided, regardless of whether they pay for it or not. Non-rivalrous consumption means that one person's consumption of the good does not reduce its availability for others.

- Diagram: A diagram might illustrate the concept of non-excludability and non-rivalrous consumption, but it's typically easier to understand through examples.

- Example: A classic example of a public good is national defense. Once a country invests in its defense, it benefits all citizens, regardless of whether they contribute taxes towards it or not. Additionally, one person's protection by the military does not diminish the protection available to others. Similarly, street lighting is a public good as it benefits everyone in the community and cannot be excluded from use by non-payers.

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Q. No. 2 Consider a Pay off Matrix of Prisoners Dilemma, given as under

Prisoner A

Prisoner B

Confess

Confess

Not Confess

A: 3 Years

A: 06 Months

B: 3 Years

B: 10 Years

Not Confess

A: 10 Years

A: 2 Years

B: 06 Months

B: 2 Years

Discuss Nash equilibrium and Dominant Strategy in this case.

Definitions:

1. Nash Equilibrium:

- Definition: Nash equilibrium is a concept in game theory where each player's strategy is optimal given the strategies chosen by the other players. In other words, no player has an incentive to unilaterally change their strategy.

2. Dominant Strategy:

- Definition: A dominant strategy is a strategy that provides the highest payoff for a player regardless of the strategy chosen by the other player(s). It is the most rational choice for a player, considering all possible scenarios.

Diagram:

In the context of the Prisoner's Dilemma, we can represent the payoff matrix with a diagram. Here's a simplified version:

yaml

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B: Confess B: Not Confess A: Confess (3, 3) (0, 10) A: Not Confess (10, 0) (2, 2)

Examples:

Let's illustrate Nash equilibrium and dominant strategy with examples:

1. Nash Equilibrium Example:

- Suppose Prisoner A and Prisoner B are arrested for a crime, and the police offer them a deal. Each prisoner can choose to confess or not confess. The payoffs in years of imprisonment are given in the payoff matrix.
- Nash equilibrium occurs when both prisoners confess. In this scenario, neither prisoner has an incentive to change their strategy. If one prisoner changes their strategy (e.g., from confess to not confess), the other prisoner's best response is still to confess. Thus, both prisoners end up confessing, resulting in a Nash equilibrium.

2. Dominant Strategy Example:

- Consider the same Prisoner's Dilemma scenario. Analyzing the payoff matrix, we find that both Prisoner A and Prisoner B have a dominant strategy to confess.
- Prisoner A's dominant strategy is to confess because regardless of Prisoner B's choice, confessing yields a higher payoff for Prisoner A (3 or 10 years vs. 0 or 2 years).

- Similarly, Prisoner B's dominant strategy is to confess because confessing always results in a higher payoff for Prisoner B compared to not confessing (3 or 0 years vs. 10 or 2 years).

Q. No. 3 State and prove the condition for Pareto optimum in exchange.

Explanation:

Pareto Optimality in Exchange:

Pareto optimality, named after the Italian economist Vilfredo Pareto, refers to a state of allocation of resources in which it is impossible to make any one individual better off without making at least one individual worse off. In the context of exchange, Pareto optimality occurs when no further exchange between individuals can increase the welfare of any individual without decreasing the welfare of another individual.

Condition for Pareto Optimality:

The condition for Pareto optimality in exchange can be stated as follows:

In an exchange economy with n individuals and m goods, an allocation of goods is Pareto optimal if and only if there are no further mutually beneficial trades possible.

Proof of Pareto Optimality:

To prove that an allocation of goods is Pareto optimal, we need to demonstrate that no further mutually beneficial trades are possible. This can be done by assuming the opposite — that there exists a trade that can make at least one individual better off without making any other individual worse off.

1. **Assume the Opposite:** Let's assume there exists a trade that can make at least one individual better off without making any other individual worse off.
2. **Examine Potential Trades:** Consider the possible trades that could be made to improve the welfare of the individual(s) without decreasing the welfare of others.
3. **Analyze the Impact:** Evaluate the impact of these potential trades on the welfare of all individuals involved. If any trade results in at least one individual being made

better off without making any other individual worse off, it contradicts the assumption of Pareto optimality.

4. **Contradiction:** Since Pareto optimality requires that no further mutually beneficial trades are possible, if any trade can improve the welfare of at least one individual without decreasing the welfare of any other individual, the initial allocation cannot be Pareto optimal.
5. **Conclusion:** Therefore, by contradiction, if no further mutually beneficial trades are possible, the initial allocation must be Pareto optimal.

Example:

Consider an economy with two individuals, Alice and Bob, and two goods, apples and oranges. If Alice has more oranges than she desires and Bob has more apples than he desires, they could potentially trade to achieve a better allocation of goods. However, if no such mutually beneficial trades are possible, the initial allocation is Pareto optimal.

